

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	A Phase 3 Study of Pembrolizumab in Combination With Carboplatin/Taxane (Paclitaxel or Nab-paclitaxel) Followed by Pembrolizumab With or Without Maintenance Olaparib in the First-line Treatment of Metastatic Squamous Non-small Cell Lung Cancer (NSCLC)
Short Title:	A Study of Pembrolizumab (MK-3475) With or Without Maintenance Olaparib in First-line Metastatic Squamous Non-small Cell Lung Cancer (NSCLC, MK-7339-008/KEYLYNK-008)
Protocol Number:	324210250010471026
NCT Number:	NCT03976362
Sponsor Name:	Merck
Investigational Product:	carboplatin
Phase:	PHASE3
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Progression-free Survival (PFS): Progression-free Survival was defined as the time from the date of randomization until either documented disease progression or death due to any cause, whichever occurred first. Per Response Evaluation Criteria in Solid Tumors (RECIST) 1.1, progressive disease (PD) was defined as ≥20% increase in the sum of diameters of target lesions. In addition to the relative increase of 20%, the sum must have also demonstrated an absolute increase of ≥5 mm. The appearance of one or more new lesions was also considered PD. PFS as assessed by Blinded Independent Central Review (BICR) per RECIST 1.1 is presented. PFS is reported based on the non-parametric Kaplan-Meier method. Per protocol this outcome measure was not planned for the Induction Phase. 2. Overall Survival (OS): Overall survival was the time from the date of randomization to death due to any cause. OS is reported based on the non-parametric Kaplan-Meier method. Per protocol this outcome measure was not planned for the Induction Phase.
Secondary Obj.:	1. Number of Participants With One or More Adverse Events (AEs) 2. Number of Participants Who Discontinued Study Intervention Due to an AE 3. Change From Baseline in European Organization for Research and Treatment of Cancer (EORTC) Quality of Life Questionnaire-Core 30 (QLQ-C30) Global Health Status (GHS)/ Quality of Life (QoL) (Items 29 and 30) Combined Scale Score

2.2 Background & Rationale

The current study will compare pembrolizumab (MK-3475) plus maintenance olaparib, vs. pembrolizumab plus maintenance olaparib placebo for the treatment of squamous NSCLC. The study's 2 primary hypotheses are:

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3. Study Design & Methodology

Design Framework: Type: INTERVENTIONAL, Phase: PHASE3, Status: ACTIVE_NOT_RECRUITING

Target N: 851

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

1. Have a histologically or cytologically confirmed diagnosis squamous NSCLC.
2. Have Stage IV squamous NSCLC.
3. Have measurable disease based on RECIST 1.1.
4. Have not received prior systemic treatment for their advanced/metastatic NSCLC.
5. Have provided archival tumor tissue sample or newly obtained core or incisional biopsy of a tumor lesion not previously irradiated.

Note: Adequacy of biopsy specimen for the above analyses must be confirmed by the central laboratory before the participant can receive study intervention(s). Submission of another tumor specimen may be required prior to enrolling the participant, if adequate tumor tissue was not provided the first time.

6. Have a performance status of 0 or 1 on the Eastern Cooperative Oncology Group (ECOG) Performance Status assessed within 7 days prior to the administration of study intervention
7. Have a life expectancy of at least 3 months.
8. Has adequate organ function.
9. Male and female participants who are not pregnant and of childbearing potential must follow contraceptive guidance during the treatment period and for 180 days afterwards.
10. Male participants must refrain from donating sperm during the treatment period and for 180 days afterwards.

4.2 Exclusion Criteria

1. Has non-squamous histology NSCLC.
2. Has a known additional malignancy that is progressing or has progressed within the past 3 years requiring active treatment.
3. Has known active central nervous system metastases and/or carcinomatous meningitis.
4. Has a known hypersensitivity to any components or excipients of carboplatin, paclitaxel or nab-paclitaxel, or olaparib.
5. Has a severe hypersensitivity (?Grade 3) to pembrolizumab and/or any of its excipients.
6. Has an active autoimmune disease that has required systemic treatment in past 2 years.
7. Has a diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy.
8. Has a known history of human immunodeficiency virus (HIV) infection, a known history of hepatitis B infection, or known active hepatitis C virus infection.
9. Has interstitial lung disease, or history of pneumonitis requiring systemic steroids for treatment.
10. Has received prior therapy with olaparib or with any other polyadenosine 5' diphosphoribose (polyADP ribose) polymerization (PARP) inhibitor.
11. Has received prior therapy with an agent directed to programmed cell death ligand 1 (PD-L1), anti PD-L2, or directed to a stimulatory or co-inhibitory T-cell receptor (e.g., cytotoxic T-lymphocyte-associated protein 4 (CTLA-4), OX-40,

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CD137).

12. Has myelodysplastic syndrome (MDS)/acute myeloid leukemia (AML) or with features suggestive of MDS/AML.

11. Study Identifiers & Governance

Primary ID: 324210250010471026

Registry Number: NCT03976362

Sponsor: Merck

Study Title: A Study of Pembrolizumab (MK-3475) With or Without Maintenance Olaparib in First-line Metastatic Squamous Non-small Cell Lung Cancer (NSCLC, MK-7339-008/KEYLYNK-008)

15. Sign-off & Approval

Principal Investigator: _____ Date: _____

Clinical Operations Lead: _____ Date: _____

Lead Statistician: _____ Date: _____